

Supplementary Materials: Supplementary Materials for “SEMTRA: Global Semantic Transition and Rough-Set Rules for Auditable Post-hoc Explainability”

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1. Reproducibility Scope and Evidence Boundary

This supplementary material documents the numerical and procedural evidence used to support the SEMTRA audit claims. The central purpose is to make the experimental record inspectable at three levels. First, the source artifacts identify how every reported value was generated or preserved. Second, the dataset-specific notes explain what each benchmark can and cannot support. Third, the figures and tables summarize the sensitivity, uncertainty, and runtime behavior that would be too dense for the main manuscript. The supplement is intentionally conservative. It does not use additional narrative to strengthen the conclusions beyond the generated evidence, and it treats low coverage, high conflict, or low fidelity as substantive findings rather than as cosmetic defects.

The evidence boundary is especially important because SEMTRA is an audit framework, not a replacement classifier. The rulebook is designed to expose the subset of a frozen model’s behavior that can be expressed through discrete semantic predicates. If the selected concept dictionary omits important visual cues, or if the neural representation does not preserve those cues in a linearly recoverable form, the rulebook should fail visibly through abstention, conflict, or low fidelity. The generated artifacts therefore record both successful extraction and diagnostic failure. This convention is used consistently for AwA2, SUN, and Derm7pt. AwA2 supports the primary closed-world semantic audit and the official xlsa17 seen/unseen semantic-transfer validation. SUN acts as a scene-understanding stress test with a much finer category structure. Derm7pt acts as a retrospective technical validation with expert checklist concepts and a locked generic image encoder, but it is not clinical validation.

The local repository contains generated CSV, JSON, $\LaTeX$, PDF, and log files under the ‘outputs’ hierarchy. Raw datasets, downloaded image archives, model weights, and local virtual environments are excluded from the shareable bundle. This separation prevents accidental redistribution of large or externally licensed data while preserving the exact derived artifacts needed for inspection. The source paths listed in Table S1 are intentionally retained as machine-readable provenance references. They are not part of the scientific prose claim; they are pointers to the reproducibility record.

Table S1. Artifact index used by the supplementary evidence record.

Location	Contents
outputs/revision_v1/awa2/	AwA2 dataset manifest, Protocol A seed-wise metrics, Protocol B seed-wise and per-class metrics, selected attributes, thresholds, rulebooks, prediction exports, and WEDD-vs-MDLP paired statistics.
outputs/revision_v1/sun/	SUN xlsa17 manifest, transition metrics, selected attributes, thresholds, rulebook, prediction exports, and q-sensitivity outputs.
outputs/revision_v1/derm7pt/	Derm7pt manifest, concept-transition metrics, rulebook, thresholds, official-split test predictions, and q-sensitivity outputs.
outputs/revision_v1/sensitivity/	AwA2 selected-attribute count, SVD-rank, and confidence-threshold sensitivity grids.
outputs/revision_v1/runtime/	Per-stage runtime JSONL logs and summarized runtime CSV.
outputs/revision_v1/tables/	L ^A T _E X-ready tables generated from machine-readable artifacts.
outputs/revision_v3/	Enhanced prediction exports with true labels and base predictions, object-level bootstrap intervals, SUN hierarchy diagnostics, locked Derm7pt ResNet-50 features, QC records, and the shareable submission bundle.

The experiments were run with fixed seeds where seed-wise variation is scientifically meaningful. For AwA2, the primary Protocol A regeneration used seeds 42, 43, 44, 45, and 46. These seeds affect the train/validation/test partitioning and the induced symbolic rulebook. The reported means and standard deviations are therefore seed-wise summaries of the complete pipeline rather than post-hoc perturbations of a single trained rulebook. For object-level uncertainty, the enhanced prediction exports under outputs/revision_v3/statistics/ support nonparametric bootstrap intervals computed directly over prediction rows. This second uncertainty source addresses a different question: how stable a reported accuracy, fidelity, coverage, or conflict estimate is over the evaluated objects for a fixed exported prediction table.

The execution environment was configured as a local Python environment with NumPy, pandas, SciPy, scikit-learn, matplotlib, pyarrow, h5py, Pillow, torch, and torchvision. The package versions recorded in the v3 manifest are the reproducibility reference for the generated outputs. The Derm7pt ResNet-50 feature extraction uses the TorchVision ImageNet-1K v2 weights in evaluation mode, with no gradient updates and no dermatology-specific training. This design makes the feature cache reproducible and traceable, but it also limits clinical interpretation. The encoder is a technical feature extractor for evaluating the SEMTRA audit sequence under a medical-concept vocabulary, not a diagnostic medical device.

2. Dataset Protocols

2.1. AwA2 and xlsa17

AwA2 is the main benchmark because it provides a mature bridge between visual representations and semantic attributes. The xlsa17 release supplies 2048-dimensional ResNet-101 feature vectors and class-level semantic prototypes for the Animals with Attributes 2 dataset [1,2]. This setting is well matched to the SEMTRA question: can a frozen representation be mapped into an interpretable semantic coordinate system and then converted into auditable symbolic rules? Protocol A evaluates this question in a closed-world setting where training, validation, and test examples come from the same class set. Protocol B follows the official seen/unseen split and is used only as semantic-transfer validation.

The AwA2 Protocol A output should be read as an audit of the model's semantic logic. The base predictor reaches 71.16% top-1 test accuracy, while the symbolic rulebook reaches 39.73% covered accuracy and 40.48% covered fidelity on covered cases. The difference is the audit tax introduced by semantic reconstruction, discretization, rough-set lower approximation, and conflict-aware inference. The tax is large, but it is not hidden. SEMTRA reports coverage, abstention, conflict, fidelity to the base predictor, and accuracy against labels so that a user can see which part of the behavior is currently expressible through simple semantic predicates.

The enhanced AwA2 prediction exports are more detailed than the summary tables. Each row contains the true class, base prediction, rulebook prediction, abstention state, conflict state, activated rule identifiers, correctness on covered cases, and fidelity to the base model on covered cases. These columns make it possible to recompute the metrics without reverse-engineering hidden arrays from the training script. They also allow object-level confidence intervals to be computed directly. Figure S1(a) summarizes this object-level audit behavior.

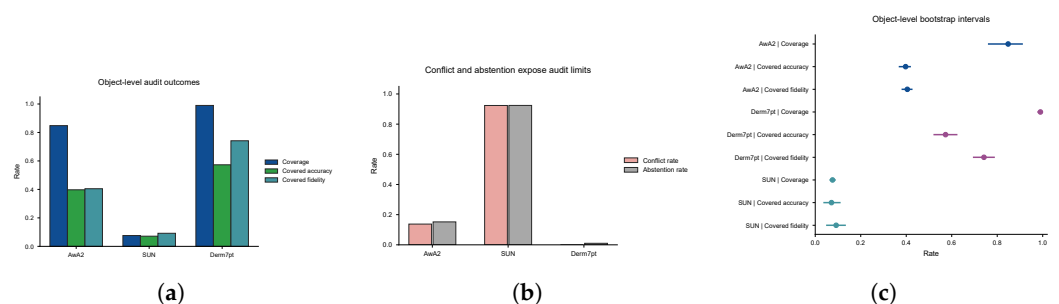


Figure S1. Object-level audit outcomes computed from enhanced prediction exports. The panels separate coverage, conflict, abstention, covered accuracy, covered fidelity, and bootstrap uncertainty so that predictive correctness and faithfulness to the frozen base model are not conflated.

The most important interpretive distinction in the AwA2 audit is between accuracy and fidelity. Covered accuracy compares the symbolic rulebook prediction with the dataset label. Covered fidelity compares the same symbolic prediction with the frozen base predictor. These values can differ because the base predictor is imperfect. If the base predictor is wrong but the rulebook predicts the label, covered accuracy increases while fidelity decreases. If the rulebook reproduces a wrong base prediction, fidelity increases while accuracy decreases. Reporting both metrics prevents a common ambiguity in post-hoc explanation papers: an explanation can be faithful to a flawed model without being correct in the dataset sense.

Conflict and abstention complete this picture. A high-conflict case is not merely a wrong prediction; it is a case where multiple symbolic explanations compete under the selected semantic representation. An abstention is not a missing output but a declared failure to verify the case through the current rulebook. Figure S1(b) shows that these quantities are not minor details. They identify the limits of the symbolic audit layer, and they give the practitioner a reason to inspect classes, attributes, or representation neighborhoods before trusting the explanation.

2.2. SUN Attribute Database

SUN is included because it tests whether the same audit protocol remains meaningful outside animal attributes. The official SUN Attribute Database contains 102 discriminative scene attributes over more than 700 categories and more than 14,000 images [3,4]. The xlsa17 SUN feature release supplies the ResNet-101 representation and split structure used

by the local runner. This setting is deliberately more difficult for a global symbolic rulebook because scene categories often differ by spatial layout, material, viewpoint, and context rather than by compact object attributes.

Table S2. SUN xlsa17 semantic-transition and rulebook audit summary.

Scope	Objects	Classes	MAE	Coverage	Covered fidelity
Unseen prototype transfer	1440	72	0.0683	–	–
Seen-test rulebook audit	2580	645	–	0.0760	0.0918

The SUN results should be interpreted as a stress test rather than as evidence of broad scene-recognition strength. The generated SUN row has low rulebook coverage and low covered fidelity. That behavior is scientifically useful because it shows that the current global symbolic template does not automatically generalize to fine-grained scene categories. A method that silently forced predictions would hide this failure. SEMTRA instead exposes the weakness through abstention and conflict. The v3 SUN metadata join further localizes this behavior by aggregating outcomes over normalized xlsa17 image paths and official SUN hierarchy categories.

Table S3. SUN v3 low-coverage category diagnostics.

Category	Objects	Coverage	Conflict	Covered fidelity
abbey	4	0.0000	1.0000	–
access_road	4	0.0000	1.0000	–
airfield	4	0.0000	1.0000	–
airplane_cabin	4	0.0000	1.0000	–
airport/airport	4	0.0000	1.0000	–
airport/entrance	4	0.0000	1.0000	–

The category diagnostics show that low coverage and high conflict are not uniformly distributed. Some scene groups are less problematic, while many fine-grained categories produce near-total abstention. This is expected when a compact global rulebook is asked to separate hundreds of scene classes with relatively broad attributes. It also suggests a practical workflow: use SEMTRA to identify category families where the current semantic dictionary is underpowered, then enrich the concept vocabulary or restrict the audit task before claiming operational scene explainability.

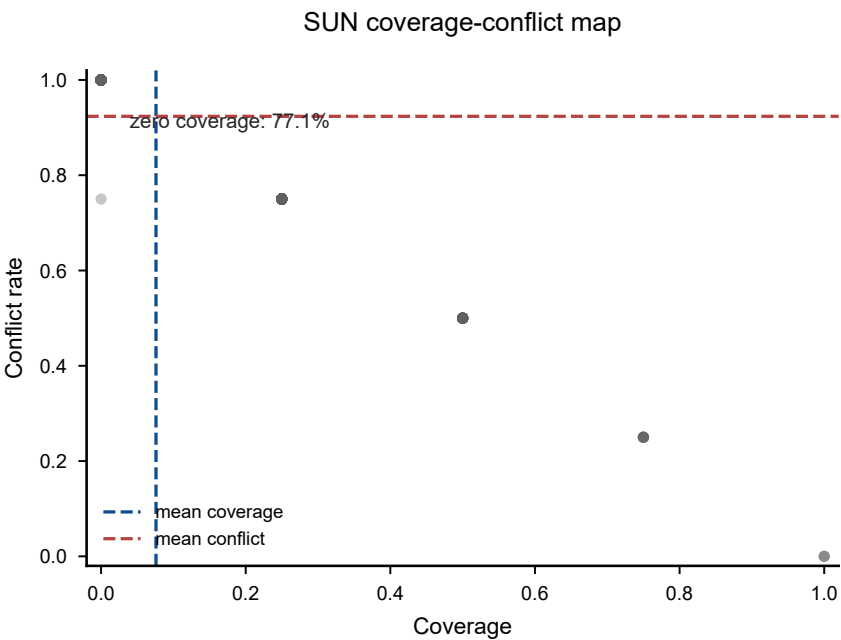


Figure S2. SUN category-level coverage and conflict map. The plot highlights the hierarchy-dependent failure modes of the current scene-audit configuration.

The SUN result is therefore not a contradiction of the AwA2 result. It shows that semantic audibility depends on concept granularity and dataset structure. AwA2 attributes are class-level animal properties, and many classes have visually grounded semantic prototypes. SUN attributes cover scene properties that can be present in multiple categories and can vary strongly within a category. Under those conditions, a single global symbolic rulebook becomes more brittle. This is a useful negative result because it prevents over-generalization of SEMTRA from animal attributes to all visual recognition tasks.

2.3. Derm7pt

Derm7pt is included to test the audit sequence under a medical concept vocabulary. The dataset provides dermoscopic and clinical imagery, structured metadata, diagnosis labels, and seven checklist concepts derived from dermatology practice [5–7]. The local evaluation uses dermoscopic images by default, official case-level train/validation/test index files, grouped diagnosis labels, and the seven checklist concepts. Missing concept values are handled explicitly before model fitting. This protocol evaluates technical traceability only; it does not evaluate clinical safety, prospective generalization, or dermatologist-level decision support.

Table S4. Derm7pt retrospective technical-validation summary using official case-level splits.

Metric	Value
Test concept MAE	0.2433
Rulebook coverage	0.9241
Covered fidelity	0.5288
Covered accuracy	0.4219
Conflict rate	0.0051

The v3 Derm7pt pipeline replaces the earlier handcrafted feature proxy with a locked ResNet-50 ImageNet-1K v2 feature encoder. This change improves provenance because the encoder, weights, preprocessing, feature dimension, number of cases, and feature hash are recorded in the generated manifest. The encoder choice is intentionally conservative: it is

widely available and reproducible, but it is not dermatology-specific. The resulting feature cache allows SEMTRA to be tested on a medical-concept benchmark without making clinical-performance claims.

Table S5. Derm7pt v3 retrospective technical-validation summary with locked ResNet-50 features.

Metric	Value
Encoder	TorchVision ResNet-50 ImageNet-1K v2
Feature dimension	2048
Test concept MAE	0.2359
Coverage	0.9899
Covered accuracy	0.5729
Covered fidelity	0.7417

Derm7pt behaves differently from SUN. Coverage is high under the locked encoder, and object-level fidelity to the base model is higher than in the SUN stress test. However, this does not imply clinical adequacy. The dataset is retrospective, the image encoder is generic, and the grouped diagnostic labels simplify the underlying dermatology task. The correct conclusion is narrower: SEMTRA can build a reproducible concept-to-rule audit layer for Derm7pt-style checklist concepts under the local protocol, and the resulting artifacts can be inspected at the diagnosis and concept levels.

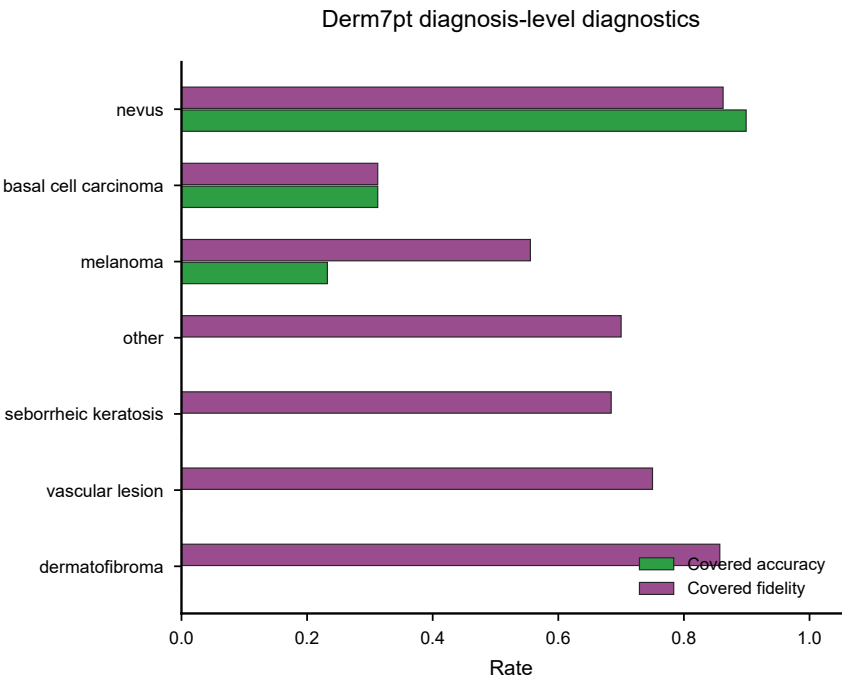


Figure S3. Derm7pt grouped-diagnosis diagnostics under the locked ResNet-50 technical-validation protocol. The figure should not be interpreted as clinical validation.

The medical interpretation boundary is worth stating explicitly. A high-fidelity rule-book can still reproduce the behavior of a non-clinical base model. A high-coverage rulebook can still miss clinically important visual patterns if those patterns are absent from the checklist concepts or are poorly encoded by the generic feature extractor. SEMTRA’s contribution in this setting is therefore auditability: it preserves a structured record of which checklist concepts activate which rules and where the symbolic layer diverges from labels or from the base predictor. Clinical use would require a domain-specific encoder,

locked preprocessing, prospective validation, calibration, subgroup analysis, and external review.

3. Discretization Evidence

The discretization module is central because every downstream rule depends on the conversion from continuous reconstructed attributes to symbolic states. WEDD combines entropy and density terms to favor thresholds that separate decision behavior while avoiding dense overlap regions. The generated comparison with equal-frequency, equal-width, and MDLP-like entropy discretization is intentionally included in the main manuscript because it directly affects the methodological claim. The supplementary record adds the paired interpretation.

The five-seed summary shows that WEDD and MDLP-like entropy are close on the primary audit metrics. WEDD has mean coverage 0.8480, covered fidelity 0.4048, covered accuracy 0.3973, and conflict rate 0.1375. The MDLP-like entropy variant has mean coverage 0.8334, covered fidelity 0.3975, covered accuracy 0.3912, and conflict rate 0.1482. The paired WEDD-minus-MDLP differences are modest: +0.0146 coverage, +0.0073 covered fidelity, +0.0061 covered accuracy, and -0.0107 conflict rate. These numbers do not support an unconditional superiority claim. They support a narrower statement: WEDD is a density-aware discretization option that performs competitively and slightly improves the chosen operating point in the regenerated AwA2 evidence.

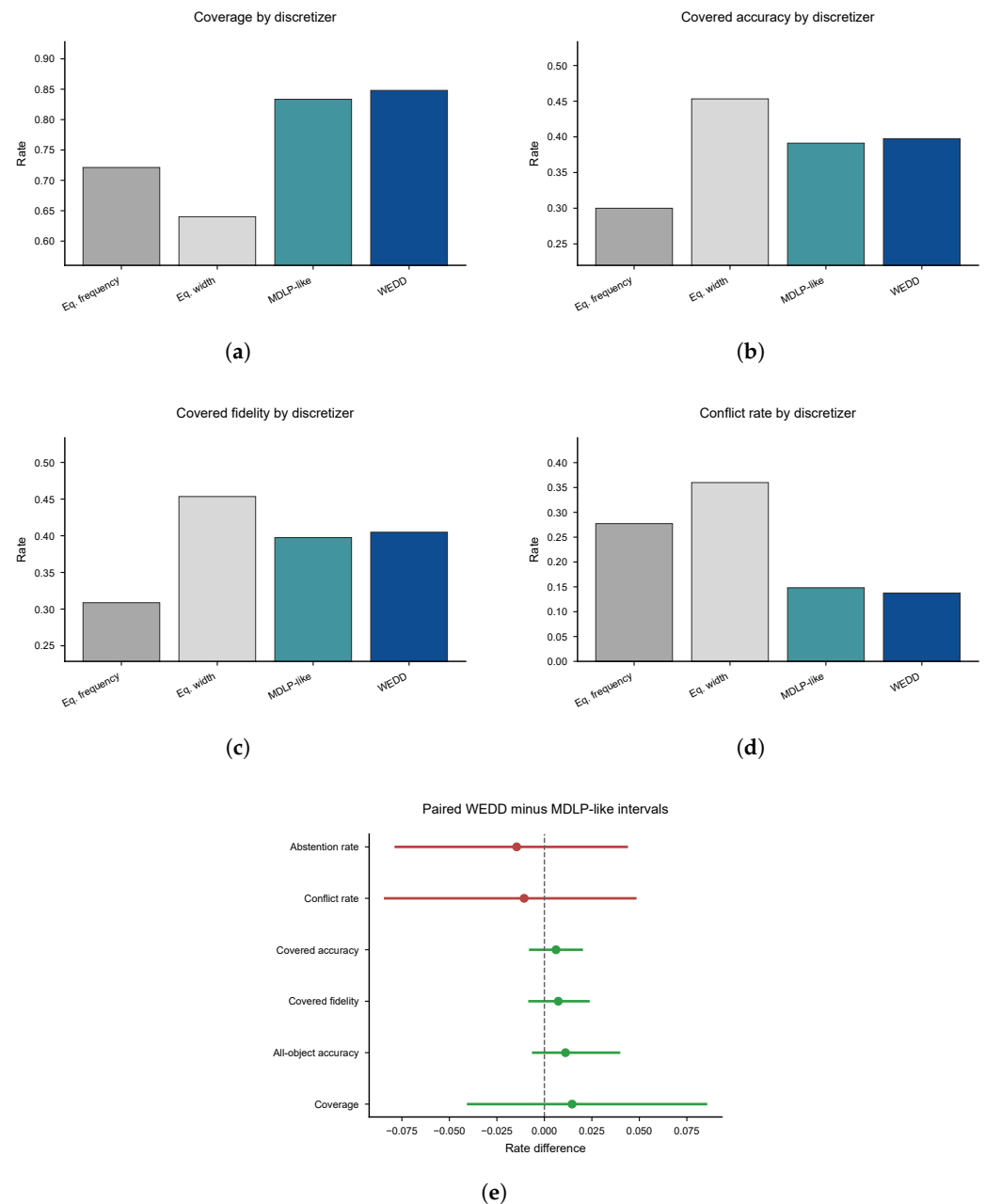


Figure S4. Discretizer evidence across coverage, accuracy, fidelity, conflict, and paired WEDD-minus-MDLP-like entropy intervals. The panels support cautious framing: the observed WEDD advantages are modest and should be treated as operating-point evidence rather than universal dominance.

This cautious reading matters because a discretizer can optimize one audit property while degrading another. Equal width produces higher covered fidelity in the regenerated comparison but has lower coverage and higher conflict. Equal frequency gives a compact distribution of states but reduces covered accuracy and fidelity. MDLP-like entropy is a strong alternative that performs close to WEDD. WEDD is retained because its threshold objective is aligned with the intended audit behavior: avoid unstable cuts through dense regions and prefer thresholds that preserve interpretable symbolic boundaries. The empirical evidence says that this choice is reasonable, not that it is mandatory.

Practitioners should therefore treat discretization as a design axis. For a dataset where false symbolic certainty is especially costly, a discretizer with lower coverage but higher fidelity may be preferable. For a dataset where broad audit coverage is needed for triage,

a WEDD-like operating point may be more useful. The repository preserves seed-wise CSV files so these tradeoffs can be recomputed without changing the manuscript narrative. This is why the claim is framed as a design-space comparison rather than a single-method victory.

4. Sensitivity to Attribute Count, Rank, and Confidence

SEMTRA has auditor-facing control knobs that affect both interpretability and performance. The number of selected attributes q controls the dimensionality of the symbolic state. The SVD rank r controls how much representational variance is preserved before semantic transition. The minimum rule-confidence threshold controls how permissive the rulebook is during induction and inference. These parameters should not be treated as arbitrary constants. They define the operating point of the audit system.

Table S6. Selected-attribute-count sensitivity on AwA2 for the WEDD rulebook.

q	Coverage	Covered fidelity	Covered accuracy	Conflict	Rules
8	0.7999	0.3435	0.3422	0.1711	59
12	0.8251	0.4002	0.3889	0.1522	60
18	0.8469	0.4109	0.4054	0.1378	57
24	0.8354	0.4343	0.4266	0.1365	54
32	0.8719	0.5522	0.5380	0.1027	50

The q -sensitivity grid shows that selected-attribute count changes the combinatorial space for reduct induction. Small q values create compact rulebooks and may improve readability, but they can omit attributes required for class separation. Large q values expose more semantic information but increase the number of possible granules and can make the reduct search more expensive. In the regenerated AwA2 sensitivity file, increasing q from 8 to 32 changes coverage, covered accuracy, covered fidelity, rule count, granule count, and threshold count. This confirms that $q = 15$ or nearby settings should be justified as operating choices rather than treated as self-evident defaults.

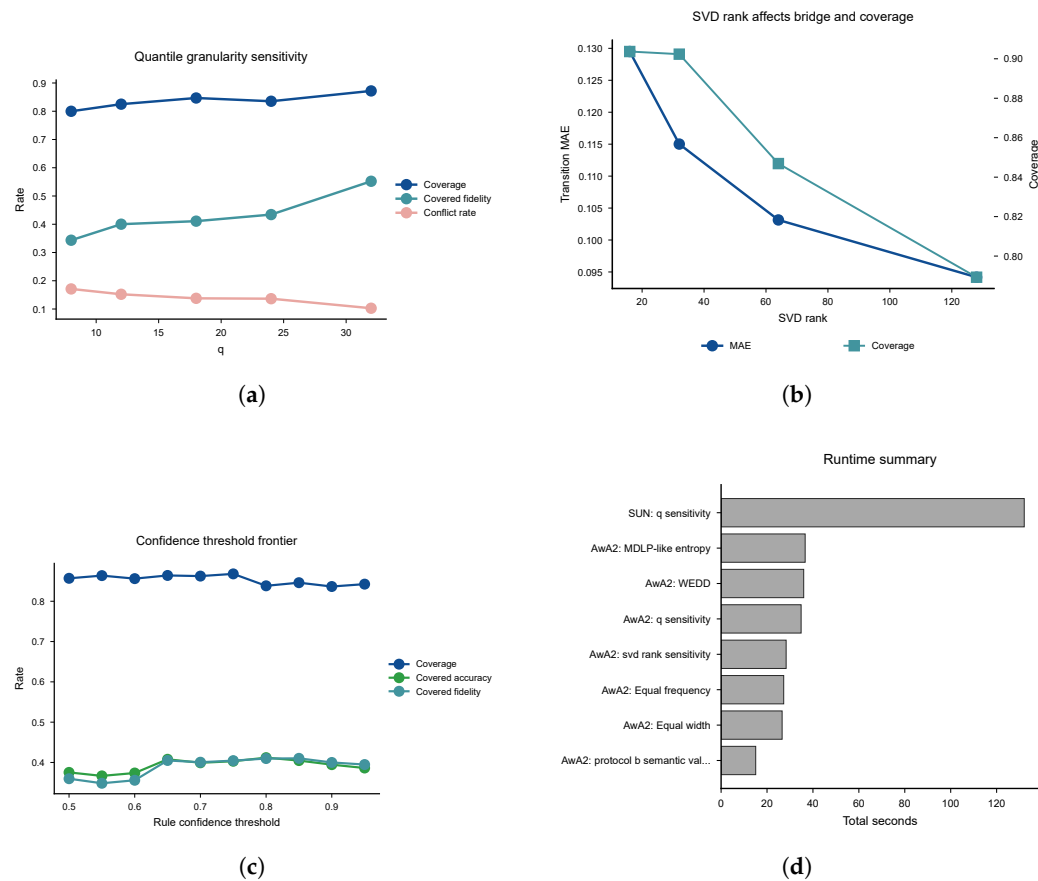


Figure S5. Sensitivity and runtime controls for the SEMTRA audit. The panels show selected-attribute count, retained SVD rank, rule-confidence threshold behavior, and phase-level runtime.

Table S7. SVD-rank sensitivity on AwA2 for semantic reconstruction and rulebook audit metrics.

Rank	MAE	Coverage	Covered fidelity	Covered accuracy	Runtime s
16	0.1295	0.9037	0.3875	0.3913	7.4
32	0.1150	0.9023	0.3668	0.3695	7.2
64	0.1031	0.8469	0.4109	0.4054	6.8
128	0.0942	0.7893	0.4898	0.4801	6.8

The SVD-rank grid addresses the audit-tax concern directly. Retaining too little rank can remove useful class evidence before the transition matrix is fitted. Retaining too much rank can preserve noise and increase downstream complexity. The regenerated grid shows that higher rank can improve covered fidelity while changing coverage and conflict behavior. This makes the audit tax tunable: a practitioner can choose more compact, stable symbolic explanations or a higher-fidelity but potentially more complex setting. The main manuscript reports this as a design decision rather than a fixed philosophical argument.

The confidence-threshold frontier is retained as a machine-readable artifact even though it is not reproduced as a central table. It is useful for post-submission auditing because it shows how coverage, covered accuracy, and covered fidelity move as rule acceptance becomes stricter. A high threshold can reduce false certainty but can also increase abstention. A lower threshold can improve coverage while permitting weaker rules. The practical choice depends on the application: documentation and exploratory debugging may tolerate lower confidence, whereas high-stakes review requires stricter thresholds and explicit abstention.

5. Runtime and Reduct Search

Table S8. Revision-v1 per-phase runtime summary from JSONL instrumentation.

Dataset	Phase	Runs	Mean s	Total s
AwA2	protocol_a_Equal frequency	5	5.5	27.3
AwA2	protocol_a_Equal width	5	5.3	26.6
AwA2	protocol_a_MDLP-like entropy	5	7.3	36.6
AwA2	protocol_a_WEDD	5	7.2	36.0
AwA2	protocol_b_semantic_validation	5	3.0	15.1
AwA2	q_sensitivity	5	7.0	34.9
AwA2	svd_rank_sensitivity	4	7.1	28.3
Derm7pt	q_sensitivity	3	0.1	0.3
Derm7pt	transition_and_rulebook	1	7.8	7.8
SUN	q_sensitivity	6	22.0	132.0
SUN	transition_and_rulebook	1	11.6	11.6

Runtime reporting is part of reproducibility because SEMTRA contains both linear algebra and symbolic search. The transition and reconstruction steps are dominated by matrix operations. WEDD thresholding requires attribute-wise scans over candidate splits. Reduct induction and rulebook construction can become expensive because the search space grows with the number of selected attributes and discrete states. The runtime logs therefore separate phases rather than reporting a single total.

The generated runtime summary shows that AwA2 Protocol A WEDD runs average about 7.19 seconds across five seeds, while the q-sensitivity grid and SVD-rank grid add additional phase-specific runtime. SUN q-sensitivity is substantially more expensive because the scene benchmark contains a larger category structure and more complex symbolic separation. Derm7pt is smaller and therefore runs quickly in the tabular rulebook phases, although the locked ResNet-50 feature extraction is a separate image-encoding cost. These distinctions matter because the practical audit tax is not only an accuracy drop. It also includes the computational cost of selecting attributes, fitting thresholds, searching reducts, and validating rule behavior.

The reduct-search timing should be read together with the q sensitivity. If a user increases the selected attribute count to improve fidelity, the symbolic search space expands. If a user lowers the confidence threshold to increase coverage, more rules may survive and later inference can become denser. These interactions are why SEMTRA reports runtime as part of the evidence record. A rulebook that is marginally more accurate but much harder to regenerate may be a poor operating point for routine audits.

6. Protocol B and Per-Class Failure Diagnosis

Protocol B evaluates semantic transfer across the official xlsa17 seen/unseen split. This experiment is not a zero-shot leaderboard claim. It asks whether the learned semantic transition retains enough class-level structure to map unseen objects toward their semantic prototypes. The result is meaningful because SEMTRA is built around semantic reconstruction. If the transition fails under seen/unseen transfer, the downstream rulebook cannot be expected to provide robust symbolic explanations for unseen classes.

Table S9. AwA2 Protocol B semantic-transfer validation across seeds. This is not a competitive zero-shot leaderboard.

Metric	Mean $\pm$ SD
Prototype unseen accuracy	0.4402 $\pm$ 0.0122
Symbolic-template unseen accuracy	0.2348 $\pm$ 0.0170
Unseen semantic MAE	0.1888 $\pm$ 0.0004

The seed-wise Protocol B summary shows moderate continuous semantic transfer and a much weaker symbolic-template result. The continuous prototype variant preserves more information because it can use distances in the reconstructed semantic space. The symbolic template variant forces the same evidence into discrete rule antecedents and therefore loses additional information. This drop is not presented as competitive zero-shot performance. It is an audit diagnostic that quantifies the information tax caused by converting continuous semantic structure into a rigid symbolic template.

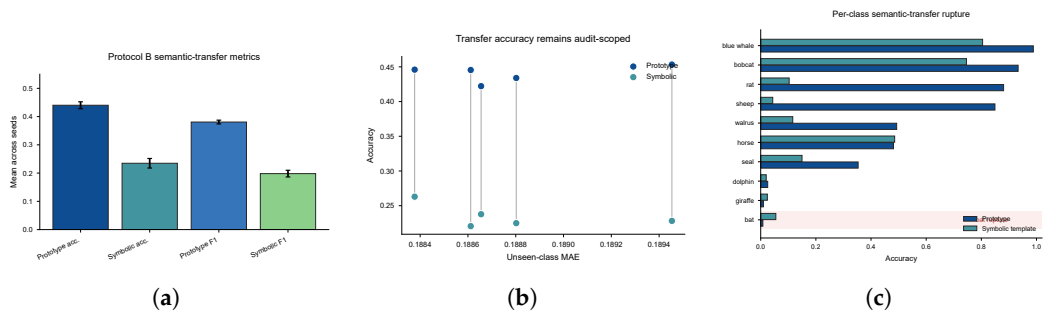


Figure S6. Protocol B semantic-transfer diagnostics. The panels summarize seed-level variation, accuracy-versus-MAE behavior, and per-class failure modes for bat, giraffe, dolphin, and sheep.

The bat failure is particularly informative. Bat has near-zero prototype accuracy but does not simply have the largest mean Hamming distance. This suggests a rupture between the ResNet-101 representation, the class-level AwA2 attribute dictionary, and the visual evidence needed for the class. Bats are visually variable, often appear in unusual poses, and depend on cues such as wings and nocturnal context that are not emphasized by the selected semantic subset. SEMTRA exposes this problem rather than smoothing it away. In a practical audit, this would motivate either a richer attribute dictionary, class-specific concept review, or a domain-specific representation before the symbolic layer is trusted.

The sheep case illustrates a different pattern. The continuous prototype score can be strong while the symbolic template collapses. That means the reconstructed semantic vector contains useful continuous information, but the discretized rule antecedents are too brittle for that class. This is exactly the kind of diagnosis that a post-hoc audit layer should provide. It separates failures of continuous semantic reconstruction from failures introduced by discretization and rule induction.

7. Cross-Domain Portability

Cross-domain evaluation is included to prevent overfitting the narrative to AwA2. The generated cross-domain table reports AwA2, Protocol B, SUN, and Derm7pt under their respective scopes. These rows are not directly interchangeable because the datasets differ in features, class structure, concept vocabulary, and target labels. The correct comparison is not "which dataset has the highest accuracy" but "what does the audit layer reveal under each semantic structure."

Table S10. Cross-domain SEMTRA revision-v1 summary. Derm7pt is retrospective technical validation only.

Dataset	Scope	Attrs	MAE	Cov.	Fid.	Acc.
AwA2	Protocol A closed-world audit	85	0.1029	0.8480	0.4048	0.3973
AwA2 Protocol B	official xlsa17 semantic-transfer validation	85	0.1888	–	–	0.4402
SUN	xlsa17 SUN unseen transfer and seen-test audit	102	0.0683	0.0760	0.0918	0.0714
Derm7pt	official test split; retrospective technical validation	7	0.2433	0.9241	0.5288	0.4219

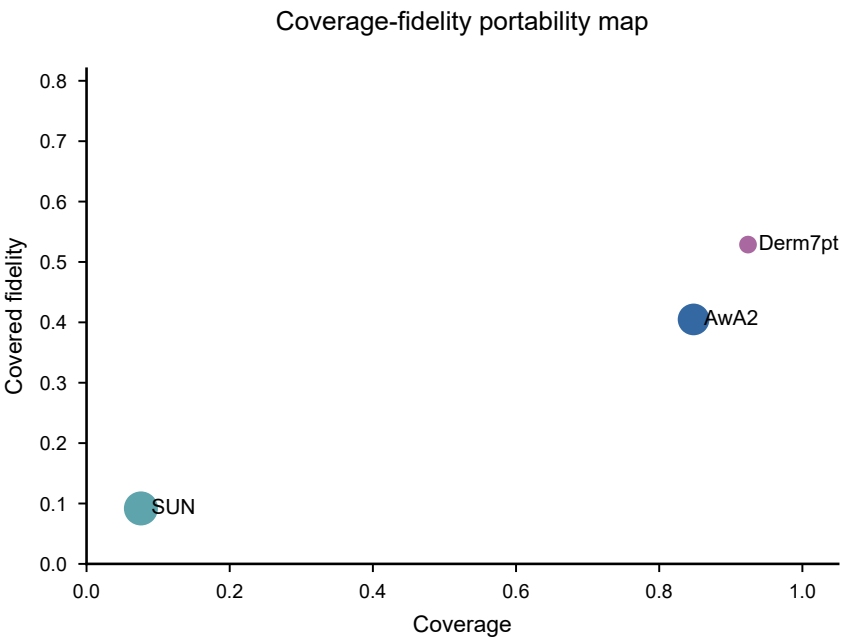


Figure S7. Cross-domain portability viewed through coverage and fidelity. The figure shows that SEMTRA’s audit behavior is strongly dataset-dependent.

AwA2 remains the strongest match to SEMTRA because the animal attributes are semantically aligned with the recognition classes and the xlsa17 feature release is designed for attribute-based transfer. SUN is a harder stress test because scene categories are numerous and semantically overlapping. Derm7pt is conceptually different again because its checklist variables are expert-defined medical descriptors, but the feature encoder used here is generic. These distinctions explain why a single summary score would be misleading. SEMTRA should be evaluated as an audit protocol whose usefulness depends on whether the concept dictionary and representation jointly support inspectable rules.

The object-level intervals generated from enhanced prediction exports add a useful layer to this cross-domain interpretation. They show how coverage, covered accuracy, fidelity, conflict, and abstention vary over the evaluated objects. For SUN, the intervals reflect a severe coverage and conflict problem. For Derm7pt, the intervals are more favorable, but the scope remains retrospective technical validation. For AwA2, the intervals quantify a stable but limited rulebook: broad coverage with modest covered accuracy and covered fidelity. These are different diagnostic outcomes, not a single monotonic ranking.

Table S11. Revision v3 object-level bootstrap intervals from enhanced prediction exports.

Dataset	Metric	Mean	95% CI	Objects
AwA2	coverage	0.8469	[0.8394, 0.8547]	7465
AwA2	covered accuracy	0.4054	[0.3937, 0.4170]	7465
AwA2	covered fidelity to base	0.4109	[0.3990, 0.4228]	7465
AwA2	coverage	0.8257	[0.8171, 0.8342]	7465
AwA2	covered accuracy	0.3824	[0.3704, 0.3942]	7465
AwA2	covered fidelity to base	0.3949	[0.3828, 0.4064]	7465
AwA2	coverage	0.8926	[0.8856, 0.8991]	7465
AwA2	covered accuracy	0.4051	[0.3937, 0.4161]	7465
AwA2	covered fidelity to base	0.4070	[0.3948, 0.4189]	7465
AwA2	coverage	0.9027	[0.8966, 0.9094]	7465
AwA2	covered accuracy	0.4048	[0.3939, 0.4172]	7465
AwA2	covered fidelity to base	0.4128	[0.4010, 0.4244]	7465
AwA2	coverage	0.7721	[0.7625, 0.7823]	7465
AwA2	covered accuracy	0.3888	[0.3765, 0.4012]	7465
AwA2	covered fidelity to base	0.3985	[0.3855, 0.4113]	7465
SUN	coverage	0.0760	[0.0659, 0.0868]	2580
SUN	covered accuracy	0.0714	[0.0389, 0.1083]	2580
SUN	covered fidelity to base	0.0918	[0.0511, 0.1313]	2580
Derm7pt	coverage	0.9899	[0.9797, 0.9975]	395
Derm7pt	covered accuracy	0.5729	[0.5229, 0.6218]	395
Derm7pt	covered fidelity to base	0.7417	[0.6964, 0.7863]	395

The interval table should be read as an object-level uncertainty summary rather than as a replacement for the seed-wise summaries. Seed-wise variation asks how much the full pipeline moves under different splits or random seeds. Object-level bootstrapping asks how much a metric would vary if the evaluated objects were resampled from the same exported prediction table. Both forms are useful. The seed-wise summaries are better for claims about reproducibility of the full training and induction process. The object-level intervals are better for auditing a fixed exported prediction artifact, because they can be recomputed directly from the row-wise prediction file without rerunning the pipeline.

For AwA2, the intervals support a stable but limited conclusion. The rulebook covers a large fraction of objects, yet covered accuracy and covered fidelity remain modest. The intervals do not erase this limitation; they quantify it. A narrow interval around a modest value is still a modest value. This distinction is important for explainability work because a stable explanation method can still be insufficient for a specific operational decision. In SEMTRA, the useful output is not the claim that every covered rule is highly accurate. The useful output is that the extent of rule-level verifiability can be measured, compared across settings, and inspected through activated rules.

For SUN, the object-level intervals reinforce the stress-test interpretation. The coverage estimate is low, conflict and abstention are high, and covered accuracy and covered fidelity have wide practical uncertainty because few objects are covered. A naive method might still output a label for each scene and report only accuracy. SEMTRA instead exposes that the current semantic-rule template is not adequate for the fine-grained SUN scene structure under the evaluated setup. This is a useful research finding because it prevents an overbroad portability claim and identifies a concrete future-work direction: richer scene attributes, hierarchical rulebooks, or category-family-specific concept dictionaries.

For Derm7pt, the intervals show that the technical-validation rulebook behaves more favorably than SUN under the locked ResNet-50 feature protocol. Nevertheless, the interval table does not convert the result into clinical evidence. The confidence interval describes a retrospective evaluation over available cases and generated predictions, not prospective deployment. The base encoder is generic, the labels are grouped, and the concept fields are tied to the dataset's retrospective annotations. The appropriate conclusion is that SEMTRA

can make the prediction and concept behavior inspectable under this controlled protocol. Any clinical interpretation would require new evidence outside the present artifact set.

The enhanced prediction exports also support manual trace inspection. An auditor can select an object with a correct covered prediction, an incorrect covered prediction, an abstention, or a conflict and then inspect the activated rules. The ‘activated_rules’ and ‘n_activated’ columns show whether a decision comes from a single rule or from multiple competing rules. The ‘faithful_to_base_when_covered’ column separates a rule that reproduces the base predictor from a rule that disagrees with it. The ‘correct_when_covered’ column separates both of those cases from the dataset label. This row-wise structure is what makes SEMTRA auditable rather than merely summarized.

8. Claim Discipline and Practical Use

SEMTRA is most useful when its output is treated as an audit artifact. A practitioner can use it to inspect which semantic attributes drive a rule, how many examples support that rule, how often the rulebook abstains, and where conflicts occur. This is valuable for model documentation, failure analysis, dataset triage, and concept-vocabulary design. It is not sufficient by itself for high-stakes deployment. In high-stakes settings, the audit layer would need to be paired with domain-specific validation, external calibration, subgroup testing, and independent review.

The acceptable audit-tax level is use-case dependent. For a screening classifier, a low-fidelity explanation layer may be unacceptable if users interpret explanations as reliable justifications. For a model-debugging workflow, the same layer may be useful because it identifies classes and semantic regions where the representation fails. For documentation, explicit abstentions may be preferable to plausible but unsupported explanations. SEMTRA’s value therefore lies in making the tradeoff explicit. It reports when the symbolic layer works, when it disagrees with the base model, and when it refuses to give a deterministic explanation.

The scope restrictions are encoded in the manuscript language and in the generated artifacts. SUN is described as a portability stress test, not a scene-recognition benchmark victory. Derm7pt is described as retrospective technical validation, not clinical deployment evidence. Protocol B is described as semantic-transfer validation, not a competitive zero-shot learning result. WEDD is described as a density-aware discretization option, not a universally superior discretizer. These restrictions are not rhetorical hedges; they are necessary conditions for making the audit record scientifically defensible.

Future work should improve both sides of the semantic bridge. On the representation side, domain-specific encoders and instance-level concept supervision could reduce the gap between latent features and human concepts. On the symbolic side, search-efficient rough-set variants, differentiable approximations, or structured pruning could reduce runtime and improve stability for large concept dictionaries. On the evaluation side, external validation should report not only accuracy but also fidelity, coverage, conflict, abstention, calibration, and subgroup behavior. The current artifact set provides the baseline against which those improvements can be measured.

A practical SEMTRA deployment should begin with a question about the intended use of explanations. If the user wants a high-accuracy classifier, the rulebook alone is not the correct artifact. If the user wants to know which part of a frozen model can be expressed through named concepts, then the rulebook, coverage, fidelity, and conflict records are directly relevant. If the user wants to debug a representation, the most informative objects may be the failures: covered disagreements, conflicts, high-distance fallback cases, and abstentions. These cases identify where the concept dictionary and the representation no longer align.

The same distinction applies to model comparison. A black-box classifier and a symbolic audit layer should not be ranked only by top-1 accuracy. The classifier is optimized to predict labels. The audit layer is constrained to use reconstructed semantic states and simple rules. The appropriate comparison asks what explanatory information is gained or lost by imposing those constraints. SEMTRA’s contribution is to quantify that loss through the audit tax and to preserve the evidence needed to decide whether the loss is acceptable for a specific task.

The generated figures are therefore not decorative. The WEDD-MDLP figure documents that the discretizer claim is modest. The q-sensitivity and SVD-rank figures show that the operating point is tunable. The runtime figure records the computational cost of the symbolic audit. The Protocol B failure figure shows where semantic transfer breaks down at the class level. The SUN and Derm7pt figures show that portability depends on dataset structure and validation scope. Together, they turn the manuscript from a single benchmark report into a documented audit protocol.

9. Reconstruction Checklist

The core reconstruction path is as follows. First, prepare the public benchmark resources outside the version-controlled repository. Second, run the provided experiment scripts to generate the AwA2, SUN, and Derm7pt outputs under the ‘outputs’ hierarchy. Third, inspect the generated manifests and runtime logs before interpreting any result. Fourth, compile the main manuscript and this supplement from the ‘manuscript’ directory. Fifth, verify that the submission bundle contains the source TeX files, references, generated tables, referenced figures, PDFs, logs, and manifests. This workflow keeps source, generated evidence, and compiled artifacts aligned.

The supplement intentionally uses generated tables and figures rather than retyping all numerical values in prose. Tables remain the authoritative compact summaries, while the CSV and JSON files remain the authoritative machine-readable sources. The prose explains how to interpret the values and where not to overstate them. This division of labor is important for a research project whose main claim is auditability. A reader should be able to trace the narrative claim to a table, a figure, a CSV file, and ultimately to a script.

The compiled PDF should be inspected together with the source bundle. A successful L^AT_EX build confirms that labels, citations, figures, and tables resolve, but it does not by itself validate the scientific scope. The manifest, QC files, and explanation record are included so that a reader can verify which artifacts were generated, which files were excluded, and which claims are intentionally limited. The final submission package should therefore contain the TeX sources, reference file, compiled PDFs, generated tables, selected figures, logs, manifests, and the Markdown explanation file. If any numerical value is changed in the manuscript, the corresponding CSV or JSON source should be updated first, then the table, figure, TeX text, and compiled PDF should be regenerated in that order.

This reconstruction discipline is part of the scientific argument. SEMTRA claims that explanations should be auditable, so the manuscript itself should be auditable. The source tree preserves that principle by separating raw data, generated evidence, compiled documents, and prose explanation. The reader can inspect the computational record without relying on hidden notebook state or unpublished interactive steps. That property is especially important for post-hoc explainability, where polished visualizations can easily overstate how much of a model’s behavior is truly understood.

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